

LAB EXERCISE 4

DEMONSTRATE HASH CRACKING USING JOHN THE RIPPER TOOL

AIM

To demonstrate password hash cracking using the John the Ripper tool on a Linux system in order to understand password vulnerabilities and the importance of strong password policies.

TOOLS / SOFTWARE REQUIRED

- Linux Operating System (Kali Linux / Ubuntu)
- John the Ripper (Jumbo version recommended)
- Terminal
- Sample password hash file
- Wordlist (rockyou.txt)

PROCEDURE

STEP 1: START LINUX SYSTEM

1. Boot into Linux (Kali Linux recommended)
2. Open the Terminal

STEP 2: VERIFY JOHN THE RIPPER INSTALLATION

Check version:

```
john
```

If not installed:

```
sudo apt update  
sudo apt install john
```

STEP 3: CREATE A SAMPLE PASSWORD HASH

Create a test password hash (MD5 example):

```
echo -n "password123" | md5sum
```

Output:

482c811da5d5b4bc6d497ffa98491e38 -

STEP 4: CREATE HASH FILE

Create a file to store the hash:

```
nano hash.txt
```

Paste the hash:

```
482c811da5d5b4bc6d497ffa98491e38
```

Save and exit (**CTRL + X, Y, ENTER**).

STEP 5: BASIC HASH CRACKING USING JOHN

Run John the Ripper:

```
john hash.txt
```

John will automatically detect the hash type and attempt to crack it.

STEP 6: VIEW CRACKED PASSWORDS

```
john --show hash.txt
```

Output:

```
482c811da5d5b4bc6d497ffa98491e38:password123
```

STEP 7: DICTIONARY ATTACK USING WORDLIST

Use rockyou wordlist:

```
john --wordlist=/usr/share/wordlists/rockyou.txt hash.txt
```

STEP 8: SPECIFY HASH FORMAT (OPTIONAL)

For MD5:

```
john --format=raw-md5 hash.txt
```

For SHA-1:

```
john --format=raw-sha1 hash.txt
```

STEP 9: CRACKING LINUX PASSWORD HASHES (SHADOW FILE – DEMO)

Combine passwd and shadow files:

```
unshadow passwd.txt shadow.txt > crack.txt
```

Run John:

```
john crack.txt
```

View results:

```
john --show crack.txt
```

STEP 10: CHECK CRACKING STATUS

```
john --status
```

STEP 11: STOP THE PROCESS

Press:

CTRL + C

OUTPUT

Password hash successfully cracked

Original plaintext password recovered

```
(kali㉿kali)-[~]
└─$ john
John the Ripper 1.9.0-jumbo-1+bleeding-aec1328d6c 2021-11-02 10:45:52 +0100 OMP [linux-gnu 64-bit x86_64 AVX2 AC]
Copyright (c) 1996-2021 by Solar Designer and others
Homepage: https://www.openwall.com/john/

Usage: john [OPTIONS] [PASSWORD-FILES]

Use --help to list all available options.

(kali㉿kali)-[~]
└─$ echo -n "password123" | md5sum
482c811da5d5b4bc6d497ffa98491e38  -

(kali㉿kali)-[~]
└─$
(kali㉿kali)-[~]
└─$ nano hash.txt
```

```
(kali㉿kali)-[~]
└─$ john --status
0g 0:00:00:21  3/3 0g/s 22013Kp/s 22013Kc/s 22013KC/s
```

RESULT

Thus, hash cracking was successfully demonstrated using the **John the Ripper** tool on a Linux system.

This experiment highlights the risks of weak passwords and emphasizes the importance of strong hashing algorithms and password complexity.